

MAT1002 Final Examination

Monday, May 13, 2024

Time: 1:30 - 4:30 PM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The maximum possible score of this examination is **145**.*
3. *There are **15** questions (with parts), which are worth 155 points in total. **This means that you do not have to answer all the questions to get the full score.***
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Write down your solutions on the **answer book**.*
6. *Show your intermediate steps **except Question 1** — answers without intermediate steps will receive minimal (or even no) marks.*
7. *Express irrational numbers in exact forms instead of decimal forms; e.g., write $\sqrt{2}$ instead of 1.414..., and write $\ln 2$ instead of 0.693....*

THIS PAGE IS LEFT BLANK INTENTIONALLY.

MAT1002 Final Exam Questions

1. [24] Short questions: no intermediate step is required.

- (i) True (T) or False (F)? If a function of two variables has a local maximum at a point, then both partial derivatives of the function at that point must be equal to zero.
- (ii) True (T) or False (F)? A continuous function of two variables attains both absolute maximum and absolute minimum values on a closed and bounded region.
- (iii) True (T) or False (F)? If $\mathbf{F}(x, y, z)$ is a continuous vector field defined on a smooth curve C with its position vector denoted by $\mathbf{r}$, then $|\int_C \mathbf{F}(x, y, z) \cdot d\mathbf{r}| \leq \int_C \|\mathbf{F}(x, y, z)\| ds$, where $\|\mathbf{F}(x, y, z)\|$ denotes the length of $\mathbf{F}(x, y, z)$.
- (iv) True (T) or False (F)? The sphere $\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ is *not* simply connected.
- (v) Find the quadratic approximation of $f(x, y) = e^x \cos y$ centered at the origin according to Taylor's formula.
- (vi) Find an equation for the plane tangent to the surface $z = \ln(x^2 + y^2)$ at the point $(1, 0, 0)$. Express your answer in the form $Ax + By + Cz = D$.
- (vii) Find the directional derivative of $f(x, y, z) = 3e^x \cos(yz)$ at the origin in the direction of $\mathbf{w}$, where $\mathbf{w} = 2\mathbf{i} + 1\mathbf{j} - 2\mathbf{k} = \langle 2, 1, -2 \rangle$.
- (viii) Let $T(x, y)$ be the temperature at the point (x, y) on the plane, and suppose that

$$\frac{\partial T}{\partial x} = 8x - 4y \quad \text{and} \quad \frac{\partial T}{\partial y} = 8y - 4x.$$

The position of a moving particle at time t is given by $x = \cos t$ and $y = \sin t$, where $t \geq 0$. Find dT/dt at time $t = \pi$.

2. [5] Determine if the following limit exists or not. If it exists, find its value; otherwise, explain why it does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 4y^2}{x^2 + 2y^2}.$$

3. [6] Consider the function

$$f(x, y) = \begin{cases} \frac{\sin(\sqrt{x^2+y^2})}{\sqrt{x^2+y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Determine whether the function is differentiable at the point $(x, y) = (0, 0)$. Briefly justify.

4. [6+6] Consider the function

$$f(x, y) = \ln \left(\sqrt{(x-a)^2 + (y-b)^2} - 1 \right).$$

Here, a and b are arbitrary real number constants.

(i) Find the domain and the range of the function, and describe the function's level curves in geometric terms.

(ii) Evaluate the integral

$$\iint_R f(x, y) dA.$$

Here, R is the region given by

$$R = \{(x, y) \mid 2 \leq (x-a)^2 + (y-b)^2 \leq 3\}.$$

5. [6] Supposing the temperature T at a point $P(x, y, z)$ in space is given by the function $T = F(x, y, z)$, where

$$F(x, y, z) = 2x - 2y + 6z + 5.$$

A particle travels along the curve C given by

$$\mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + te^{-t}\mathbf{k}, \quad t \geq 0.$$

Here, t is the time and $\mathbf{r}(t)$ is the position vector at time t . Supposing $s(t)$ is the arc length function, i.e., the distance traveled by the particle along curve C over the time interval $[0, t]$. Determine the rate of change of temperature T with respect s , i.e., $\frac{dT}{ds}$, at time $t = 1/2$.

6. [9] Find the maximum and minimum values of $f(x, y, z) = x - 2y + 5z$ on the sphere $x^2 + y^2 + z^2 = 30$.

7. [9] Find the (x, y) -positions of all the local maxima, local minima, and saddle points of the function $f(x, y) = x^3 + 3xy^2 - 15x + y^3 - 15y$.

8. [6+6] Evaluate the following integrals.

(i) $\iint_D \frac{x}{1+y^2} dA$, where $D = \{(x, y) \mid x \in [0, 2], y \in [-1, 1]\}$.

(ii) $\int_0^1 \int_x^{\sqrt{x}} \frac{\sin y}{y} dy dx$.

9. [3+3+6] Consider the graph of all points on the (x, y) -plane satisfying the following equation:

$$(x^2 + y^2)^2 = xy.$$

(i) Rewrite the equation in a polar coordinate form.

(ii) Sketch the graph of the equation on the (x, y) -plane.

(iii) Find the area enclosed by the portion of the graph in the first quadrant of the (x, y) -plane.

10. [3+3+3+3] Let D be the region in the first octant (i.e., the octant given by $x \geq 0$, $y \geq 0$, and $z \geq 0$) bounded by the coordinate planes, the plane $x + y = 4$, and the cylinder $y^2 + 4z^2 = 16$. Rewrite the triple integral $\iiint_D dV$ as an equivalent iterated integral in the following orders. Then compute the value (using one of the orders).

(i) $dx dy dz$.

(ii) $dz dx dy$.

(iii) $dy dz dx$.

11. [4+4+3] Consider the vector field

$$\mathbf{F}(x, y, z) = \left\langle 12x - 5z^2, \ln(1 + z^2), \frac{2yz}{1 + z^2} - 10xz \right\rangle.$$

(i) Show that $\mathbf{F}$ is conservative on $\mathbb{R}^3$.

(ii) Find one potential function f of $\mathbf{F}$.

(iii) At the point $(x, y, z) = (0, 1, 1)$, in what unit direction does the function f in part (ii) increase the fastest?

12. [6+6] Let $\mathbf{F}(x, y) = \langle xy^2 + 2x, 4x + 1 \rangle$, and let C be the boundary of the triangle whose vertices are $(-3, 0)$, $(0, 0)$, and $(0, 3)$.

- (i) Compute the circulation of $\mathbf{F}$ along C in the *counterclockwise* direction.
- (ii) Compute the *outward* flux of $\mathbf{F}$ through C .

13. [7] Consider the vector field $\mathbf{F} = \text{curl } \mathbf{A} = \nabla \times \mathbf{A}$, where

$$\mathbf{A} = \mathbf{A}(x, y, z) = \langle 2z, 3x, 5y \rangle.$$

Compute the downward flux of $\mathbf{F}$ across the surface S given by

$$z = x^2 + y^2, \quad z \leq 4.$$

14. [3+7] Consider the vector field

$$\mathbf{F}(x, y, z) = \langle x(x^2 + y^2 + z^2), y(x^2 + y^2 + z^2), z(x^2 + y^2 + z^2) \rangle.$$

- (i) Compute $\text{div}(\mathbf{F})$.
- (ii) Let E be the intersection of the solid ball $x^2 + y^2 + z^2 \leq 1$ and the solid cone $z \geq \sqrt{x^2 + y^2}$. Let S be the boundary surface of E . Compute the outward flux of $\mathbf{F}$ across S .

15. [4+4] Let $D = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 \right\}$ with a, b, c positive.

- (i) Show that

$$\iiint_D \frac{x^2}{a^2} dV = \iiint_D \frac{y^2}{b^2} dV = \iiint_D \frac{z^2}{c^2} dV.$$

- (ii) Compute the following triple integrals

$$\iiint_D \frac{x^2}{a^2} dV \quad \text{and} \quad \iiint_D \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dV$$